Chapter V
Enterprise Resource Planning Systems:
Effects and Strategic Perspectives in Organizations

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ABSTRACT

In the age of globalization, organizations all over the world are giving more significance to strategy and planning to get an edge in the competition. This chapter discusses the Enterprise Resurce Planning (ERP) systems effects and strategic perspectives in organizations. These are significant how information technology and ERP together facilitate in aligning the business in such a way so that it should lead to excellent productivity. It further explores in what ways effects of ERP system in organizations can provide sustained competitive advantage.

INTRODUCTION

Enterprise Resource Planning (ERP) software is one of the fastest growing segments of business computing today (Luo and Strong, 2004) and ERPs are one of the most significant business software investments being made in this new era (Beard and Sumner, 2004). Davenport (1998) has declared that 'the business worlds's embrace of enterprise systems may in fact be the most important development in the corporate use of information technology in the 1990's. Mabert et al. (2001) noted that industry reports suggests as many as 30,000 companies worldwide have implemented ERP systems. According to a report by Advanced Manufacturing Research, the ERP software market is expected to grow from $ 21 billion in 2002 to $ 31 billion in 2006 and the entire enterprise applications market which includes Customer Relationship Management and Supply Chain Management software will top $ 70 billion (AM Research, 2002). Further, AMR Research

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has projected as much as $ 10 billion in global investments in ERP (as cited in Kalling, 2003). The ERP market is projected to grow from a current $15 billion to $ 50 billion in the next five years and to reach $ 1 trillion by 2010 (Bingi et al., 1999). ERP systems offer the advantage of providing organizations with a single, integrated software system linking the core business activities such as oprations, manufacturing, sales, accounting, human resources, and inventory control (Lee and Lee, 2000; Newell et al., 2003; Shanks and Seddon, 2000). According to Brown and Vessey (2003) this integrated perspective may be the first true organization-wide view available to management. According to Lee and Myers (2004) much of the literature on ERP implementation suggests that ERP systems should support the strategic objectives of the organization. They observed that some ERP vendors tend to assume that implementing their products is a straightforward translation from strategy to IT-enabled business processes.

ERP helps organizations to meet the challenges of globalization with a comprehensive, integrated application suite that includes next-generation analytics, human capital management, financials, operations, and corporate services. With support for industry-specific best practices, ERP helps organizations improve productivity, sense and respond to market changes, and implement new business strategies to develop and maintain a competitive edge. ERP is designed to help businesses succeed in the global marketplace by supporting international legal and financial compliance issues and enabling organizations to adapt internal operations and business processes to meet country-specific needs. As a result, organizations can focus on improving productivity and serving their customers instead of struggling to ensure they are in compliance with business and legal requirements around the world. Companies that automate and streamline workflows across multiple sites (including suppliers, partners, and manufacturing sites) produced 66% more improvement in reducing total time from order to delivery, according to Aberdeen's 2007 study of the role of ERP in globalization. Those companies that coordinate and collaborate between multiple sites, operating as a vertically integrated organization, have achieved more than a 10% gain in global market share. The majority of companies studied (79%) view global markets as a growth opportunity, but of those companies, half are also feeling pressures to reduce costs (Jutras, 2007). Those companies that coordinate and collaborate between multiple sites, operating as a vertically integrated organization, have achieved more than a 10% gain in global market share (Marketwire, 2007).

INFORMATION TECHNOLOGY AND STRATEGIES

Inspite of lots of literature and guidance available less than 10% of strategies effectively formulated are effectively executed (Kaplan and Norton, 1996). Mintzberg (1992) defined strategy as " a plan –some sort of consciously intended course of action, a guideline (or set of guidelines) to deal with a situation. He further tried to define strategy from the perspectives of being a plan, a ploy, a position, a pattern, and a perspective (Ikavalko, and Aaltonen, 2001). Michael Porter's (1996) definition of strategy focuses more on the outcome, "the creation of a unique and valuable position, involving a different set of activities". He believes that a strategy is a way an organization seeks to achieve its vision and mission and that a successful strategy allows a company to capture and sustain a competitive advantage. Porter's (1985) value chain methodology identified five key forces that impact on an organization's competitive position:

- The bargaining power of suppliers;
- The bargaining power of buyer;

- The threat of new entrants;
- The threat of substitute products; and
- Rivalry among existing organizations.

He believed that the impact of these forces could be influenced by strategies that focused on low cost provider and on product differentiation. These strategies determine how various discrete activities are performed to add value and eventually a competitive advantage. An organization's activities can be categorised into primary activities that can directly add value and supporting activities. These interrelated activities make up Porter's Value Chain. One of the supporting activities he identified was the technology development. Porter and Miller (1985) proposed an information intensity matrix to assist in identifying where information technology could be used strategically in the Value Chain. Later on Somogyi and Galliers (1987) supported this by identifying how information technology could be used to assist organizations to accomplish competitive advantage in the various focuses across the Value Chain. During last two decades organizations have already identified the significance of information technology in achievement of strategic objectives. Scott Morton (1991) identified five interrelated factors (Structure, Strategy, Technology, Management Processes, Individual & Roles) that influence the attainment of strategic objectives. One of these factors was information technology. A recent survey of more than 300 CEO's and CIO's identified the alignment of IT and business strategy as their number one priority (Beal, 2003). Tallon and Kraemer (2003) in a survey of 63 companies found there was significant value gained from the alignment of these strategies. Broadbent and Weill (1993) define this IT-business alignment as the extent to which business strategies were enabled, supported, and stimulated by information strategies. Teo and King (1997) proposed four different scenarios or degrees of integration between the business and IT strategies. These included:

- **Administrative integration:** Where there is little relationship between the business and IT strategy.
- **Sequential integration:** Where the business strategy is developed in firstly in isolation to the IT strategy. The IT strategy is then developed to support the business strategy.
- **Reciprocal integration:** This is where a reciprocal and interdependent relationship exists between both strategies. The IT strategy is used to support the influence business strategy.
- **Full integration:** Occurs when both strategies are developed concurrently in an integrated manner.

This increased need for closer alignment has resulted in companies focussing on Strategic Information Systems Planning (SISP) and the development of methodologies to support this (Pant and Hsu, 1995; Hackney et al, 2000). Hackney et al. (2000) identified the assumptions which underlie SISP and discuss their validity. According to them main assumption are:

- Business srategies must exist as a precursor to SISP,
- Business strategies are different from IT,
- IT and Business strategies can be aligned.

They further argue that as business strategies evolve, it is often difficult for IT strategies to respond.

ERP AND CORPORATE STRATEGIES

ERP systems are widely adopted in a diverse range of organizations and define the business model on which they operate. For many organizations they were relieved that ERP system could help them define a business srategy and provide the

IT infrastructure to support it (Davenport, 2000). Hackney et al. (2000) believe that ERP systems can provide a "dynamic stability" to the alignment of business and IT strategies. These systems can provide a stable predictable environment of which their usage can evolve in accordance with a company's business strategy. Brancheau et al. (1996) observed that alignment of an informaion system (IS) with the strategic goals and operational objectives of an organization has been an important issue through the 1980s and 1990s. ERP research has exlpored how these types of systems contribute value to an organization (Markus and Tanis, 1999; Ross and Vitale, 2000; Somers and Nelson, 2001), as well as how they should be integrated with already-existing IT resources (Hayman, 2000). Large and small organization's continue to invest between $ 300,000 and hundreds of millions of dollars in ERP software and accompanying hardware (Markus, 1999). Different business justifications, including improved productivity, reduced costs, greater operational efficiency, enhanced customer relationship management, and better supply chain management (Brown and Vessey, 2003; Mabert et al, 2001). Finally, for the return on investment in ERP systems to be achieved, these systems should yield a strategic advantage. Kalling (2003) argues whether ERP systems provide a competitive advantage. Current perceptions of ERP systems, as evidenced in trade publications and the academic literature, emphasize their role in enhancing economic efficiency and improving financial performance (Dillard and Yuthas, 2006).

Beard and Sumner (2004) observed that regarding competitive advantage one challenge is that ERP systems impose a 'common systems' approach by establishing a common set of applications supporting business operations. Successful implementation of an ERP system requires re-engineering business processes to better align with the ERP software, so that the common systems approach is imposed (Brown and Vessey, 2003; Dahlen and Elfsson, 1999). This 'common' structure approach allows for faster implementation of the ERP system because there are fewer customized pieces to the software and the limited customization means that it will be simpler to upgrade the ERP software as new versions and features emerge over time (Beard and Sumner, 2004). Another challenge related with accomplishing a competitive advantage through ERP is the significant complexity of the implementation and integration process as it often takes several years to fully implement the ERP system. This includes integrating ERP with already-existing IS and accomplishing the related reengineering of the organization (Beard and Sumner, 2004). It is time consuming process to refine the alignment of organization to the ERP system and to more fully leverage the opportunities offered by the ERP system .

As ERP provides both tangible and intangible benefits, many organizations consider them as essential information system infrastructure to be competitive in today's business world and provide a foundation for future growth. A recent survey of 800 top U.S. companies showed that ERP systems accounted for 43% of these companies budgets (Somer & Nelson, 2001). The market segment of ERP systems varies considerably from industry to industry. According to Computer Economics Inc. Report 76% of manufacturers, 35% of insurance and health companies, and 24% of Federal Government agencies already have an ERP system or are in the process of installing one (Stedman, 1999). The global market for ERP software, which was $ 16.6 billion in 1998, is expected to have had 300 billion dollars spent over last decade (Carlino, 1999). There are number of vendors of ERP systems in the market although major one is SAP with approximately 56% of the market. In Australia 9 out of the top 12 IT users were SAP customers and 45% of the total list were also SAP users (BRW, 2002). Growth in ERP systems is due to several factors for example; the need to streamline and improve business processes, better manage information system expenditure, compe-

titive pressures to become a low cost producer, increased responsiveness to customers and their needs, integrate business processes, provide a common platform and better data visibility, and as a strategic tool for the move towards e-commerce (Davenport et al. 2003; Markus et al., 2001; Somer et al., 2001).

ERP SYSTEM AND STRATEGIC ADVANTAGE

Davenport (2000) believes that ERP systems by their very nature impact on a organization's strategy, organization and culture. Mabert et al. (2001) report that companies view the standardization and integration of business processes across the enterprise as a key benefit. As one of the executives mentioned 'ERP is the digital nerve system that connects the processes across the organization and transmits the impact of an event happening in one part to the rest accurately' (Mabert et al. , 2001). Regarding tangible benefits, the companies often observed reported lower inventories, shorter delivery cycles, and shorter financial closing cycles although, the ERP system did not lead to reductions in work force or savings in operational costs in the short term (Beard and Sumner, 2004). According to Davenport (2000) cycle time reduction (e.g. cost and time reductions in key business processes), faster information transactions (e.g. faster credit checks), better financial management (e.g. shorter financial closing cycle, improved management reporting), and laying the ground work for electronic commerce (e.g. providing the back office functions for Web-based product ordering, tracking, and delivery processes), as well as a better understanding of key business processes and decision rules are general benefits of ERP systems. Laughlin (1999) suggests that the business justification for ERP includes both hard dollar savings (e.g. reductions in procurement cost, inventory, transportation, increased manufacturing throughput, and productivity) and soft dollar savings (e.g. revenue growth, margin enhancement, and sales improvements). Interestingly both Mabert et al. (2001) and Laughlin (1999) observed that there is no evidence of ERP providing headcount reduction.

When an ERP works well it can 'speed up business processes, reduce costs, increase selling opportunities, improve quality and customer satisfaction, and measure results continuously (Piturro, 1999). Regarding business benefits of ERP, managers mention many productivity enhancements, including the ability to calculate new prices instantly, more accurate cost comparisons among different facilities, better electronic data interchange with vendors and suppliers, improved forecasting, and the elimination of bottlenecks and duplicative procedures (Plotkin, 1999). Other benefits deal with eliminating the redundancies associated with leagcy systems (Beard and Sumner, 2004). For example, at Owens Corning, there were 200 legacy systems, most running in isolation from one another. Eastman Kodak had 2600 different software applications (Palaniswamy and Frank, 2000). Organizations with ERP systems have made improvements in cross functional coordination and business performance (Oliver, 1999).

According to Markus et al. (2000) larger value of ERP is measured when the organization captures actual business results (for example reduced inventory costs), but these results don't occur until the phase in which the systems have already been successfully implemented and integrated into business operations-in referred by Markus et al. (2000) as 'onward and upward phase'- a stage three evolution. Holland and Light (2001) also support that the business benefits of ERP occur in a 'third stage' of evolution, during which innovative business processes are thoroughly implemented. As observed by Beard and Sumner (2004) that ERP may not necessarily directly provide organizations with a competitive advantage through reduction of these organizations cost below or by increasing these organizations' revenues above what

would have been the case if these systems had not been implemented. They further argued that the advantages mentioned are largely value-added measures, such as increased information, faster processing, more timely and accurate transactions, and better decision-making.

CONCLUSION AND FUTURE TRENDS

ERP systems are increasingly popular IT platform that are being installed to assist organizations in better capturing, managing, and distributing organization-wide operational data to decision makers throughout the organization (Beard and Sumner, 2004). They further discussed that

Table 1. Benefits of ERP

Company	Business benefits	Source
Fujitsu	90% reduction in cycle time for quotation from 20 to 2 days 50% reduction in financial closing times from 10 to 5 days	Jensen and Johnson (1999)
Boeing	Simplification of processes	Jensen and Johnson (1999)
Pacific coast feather company	Inventory reduction; improved Customer service	Jensen and Johnson (1999)
IBM storage products company	Time for checking customer credit upon receiving an order was reduced from 15 to 20 minutes to instantaneously Responses to customer billing inquiries occurred in real time (vs. 15-20min) Entering pricing data into the system took 5 min. where it took 8 days beforehand Shipping repair and replacement was done in 3 days, compared to as many as 44 days	Jensen and Johnson (1999)
Earthgrains	On-time product delivery rate increased to 99% Operating margins improved from 2.4% to 3.9%	Bingi et al. (1999)
Par industries	Delivery performance improved from 80% on-time to more than 95%; Lead times to customers were reduced from 6 to 2 weeks Repair parts were reduced from 2 weeks to 2 days; Work-in-process inventory dropped almost 60% Life of a shop order dropped from weeks to hours	Bingi et al. (1999)

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Table 1. continued

Owens corning	Inventory levels were reduced significantly Lot sizes and machine allocations more efficient Growth in inter-facility coordination	Palaniswami & Frank (2000)
Viskase	Reductions in lead time and inventory Reduction in headcount Integration of information Decision-making times are reduced significantly Production-based decisions are tied to sales-based decisions in a timely manner	Palaniswami & Frank (2000)
Diebold	Real-time access to data across the organization Better control of manufacturing processes	Palaniswami & Frank (2000)
Valenite	Lower levels of inventory Improved customer satisfaction Faster, more accurate order processing Accurate and timely financial information	Palaniswami & Frank (2000)
Leeson	Reduction in paperwork related to order processing Ability of end-users to gain access to information Improved accuracy of financial and inventory transactions Improved currency of manufacturing databases	Palaniswami & Frank (2000)

sustainable competitive advantage with an ERP system, important characteristics that will create sustained competitive advantage and reengineering the organization without disrupting or destroying the characterisctics that gave it a competitive advantage are further research areas to know more about effects and strategic advantages of ERP. These may be include empirical, qualitative or both the approaches in the organization. Comparison of ERP implementation effects and strategic advantages in two organizations may be an interesting area. Regarding limitation and concern as observed by Lee and Myers (2004) is that ERP systems are so large and complex and take years to implement, the inclusion of today's strategic choices into the enterprise systems may significantly constrain future action. By the time the implementation of an ERP system is finished, the strategic context of the firm may have changed.

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KEY TERMS

Business Benefits:

- Also known as: business benefits of ERP
- Similar to: Combined tangible and intangible benefits to organizations due to ERP implementation
- Associated in the manuscript with: Effects and benefits of ERP implementation

Business Processes:

- Also known as: IT-enabled business processes
- Similar to: integration of business processes
- Associated in the manuscript with: business processes to better align with the ERP software

Competitive Advantage:

- Also known as: Organization's sustains profits that exceed the average said to possess competitive advantage
- Similar to: A competitive advantage is an advantage over competitiors by offering consumers greater value either by means of lower prices or by providing greater benefits
- Associated in the manuscript with: competitive advantage through ERP

ERP:

- Also known as: Enterprise Resource Planning
- Similar to: systems attempt to integrate all data and processes of an organization into a unified system
- Associated in the manuscript with: Effects and Strategic Perspectives in Organizations

Global:

- Also known as: globalization
- Similar to: Multinational Organizations, Organizations distributed at international level
- Associated in the manuscript with: global market for ERP software, global marketplace

Integrated:

- Also known as: Combining different functional aspects of the organization
- Similar to: integrated application suite
- Associated in the manuscript with: integrated software system linking the core business activities such as oprations, manufacturing, sales, accounting, human resources, and inventory control